Dietary fat and semen quality among men attending a fertility clinic

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BACKGROUND The objective of this study was to examine the relation between dietary fats and semen quality parameters. METHODS Data from 99 men with complete dietary and semen quality data were analyzed. Fatty acid levels in sperm and seminal plasma were measured using gas chromatography in a subgroup of men (n = 23). Linear regression was used to determine associations while adjusting for potential confounders. RESULTS Men were primarily Caucasian (89%) with a mean (SD) age of 36.4 (5.3) years; 71% were overweight or obese; and 67% were never smokers. Higher total fat intake was negatively related to total sperm count and concentration. Men in the highest third of total fat intake had 43% (95% confidence interval (CI): 62–14%) lower total sperm count and 38% (95% CI: 58–10%) lower sperm concentration than men in the lowest third (Ptrend = 0.01). This association was driven by intake of saturated fats. Levels of saturated fatty acids in sperm were also negatively related to sperm concentration (r= –0.53), but saturated fat intake was unrelated to sperm levels (r = 0.09). Higher intake of omega-3 polyunsaturated fats was related to a more favorable sperm morphology. Men in the highest third of omega-3 fatty acids had 1.9% (0.4–3.5%) higher normal morphology than men in the lowest third (Ptrend = 0.02). CONCLUSIONS In this preliminary cross-sectional study, high intake of saturated fats was negatively related to sperm concentration whereas higher intake of omega-3 fats was positively related to sperm morphology. Further, studies with larger samples are now required to confirm these findings.